The Hardware Bridge

Bridging Classical Electronics and Cryogenic Quantum Control Systems

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Abstract

Superconducting quantum computers work at about 10 mK, yet are controlled by electronics at room temperature. This poster reviews the **hardware bridge** between the two: cables, attenuators, filters, amplifiers and emerging cryogenic control chips. We show how signals are carried and protected from noise at each temperature stage, use thermal-photon statistics to quantify why filtering is essential, and compare approaches (cryo-CMOS, SFQ logic, multiplexing) for scaling to larger processors.

1. Introduction

A superconducting qubit has a transition frequency of about **4–8 GHz** and must stay at **10–20 mK** inside a dilution refrigerator. Every gate and readout, however, is generated by classical hardware at **300 K**: arbitrary waveform generators (AWG), mixers, digitisers and FPGAs.

The bridge must deliver ns-precise microwave pulses and read out very weak (~fW) signals without letting heat or noise reach the qubits.

2. Why the bridge is hard

- **Heat load:** cooling power is only ~10–20 µW at the mixing chamber and ~1–2 W at 4 K; every cable leaks heat.
- **Thermal noise:** 300 K Johnson noise would dephase the qubit unless attenuated and filtered.
- **Wiring:** several coax lines per qubit do not scale to 1000+ qubits.
- **Latency:** error-correction feedback needs readout, decision and pulse within ~1 µs.

3. Temperature stages of the bridge

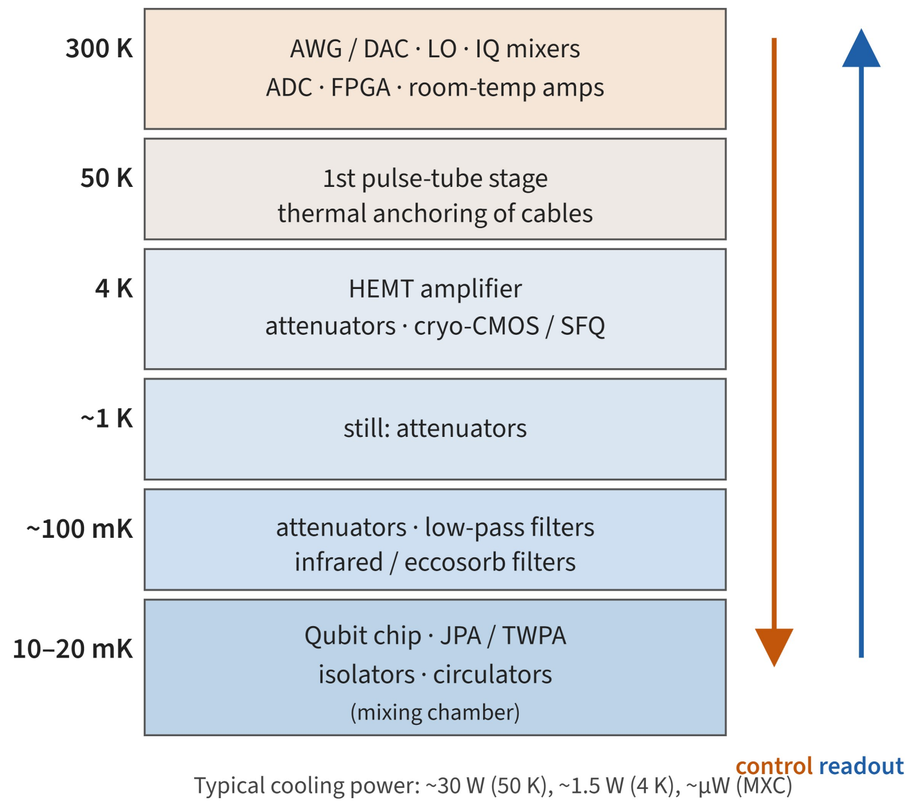


Fig. 1: Control pulses travel down, readout signals travel up. Values are typical, not device-specific.

4. Signal chain: control (down) and readout (up)

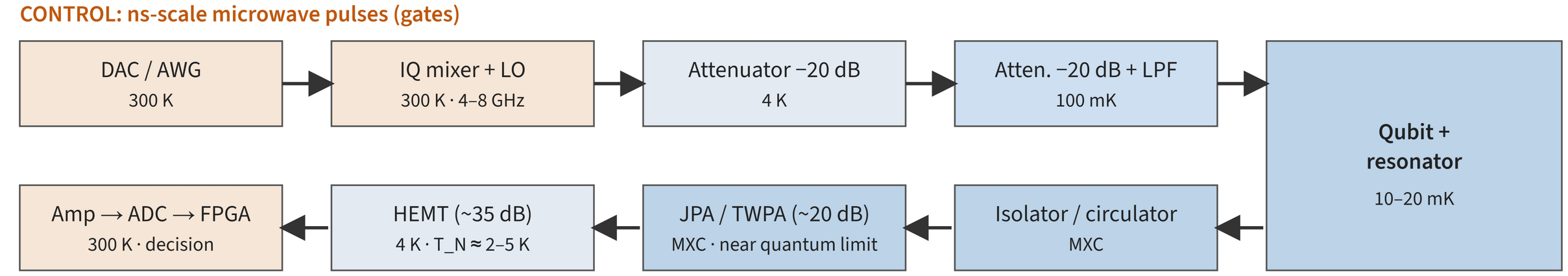


Fig. 2: Attenuation (~60 dB total) thermalises the line; amplification restores a ~fW readout signal above noise.

5. Results: thermal photons vs temperature

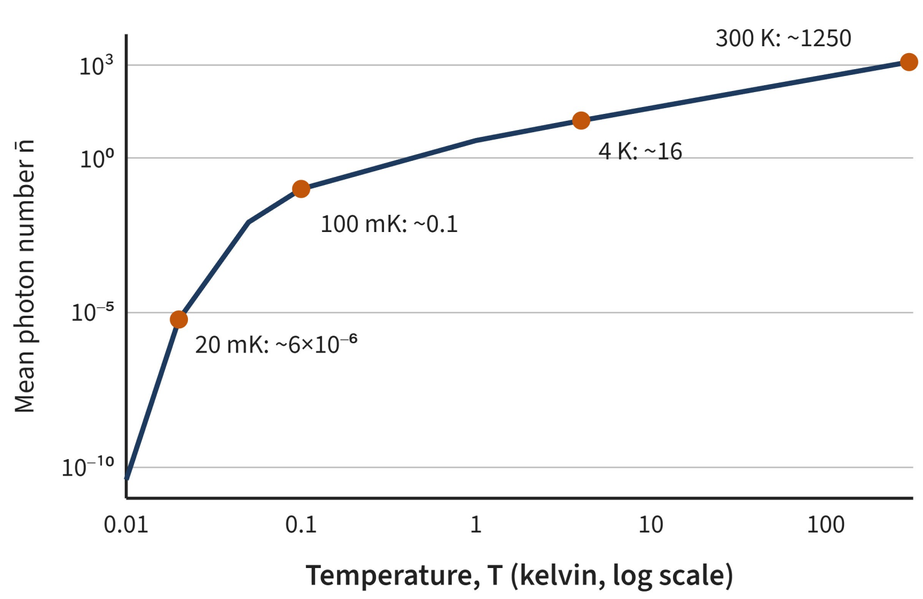


Fig. 3: Bose–Einstein occupation $\bar{n} = 1/(e^{hf/kT} - 1)$ at f = 5 GHz (hf/k ≈ 0.24 K). Qubits need $\bar{n} \ll 1$.

6. Discussion: approaches to scaling

Approach	Strength	Limitation
Room-temp electronics + coax	Flexible, low noise, mature, easy to reprogram	Wiring and heat load grow with qubit count
Cryo-CMOS at 4 K	Compact, many channels per chip, shorter latency	Power budget (~mW/qubit); device behaviour shifts when cold
SFQ logic at 4 K	Very low switching energy, fast pulses	Specialised fabrication, limited circuit complexity
Multiplexed readout	Many resonators on one feedline	Crosstalk, frequency crowding

Going from 300 K to 20 mK cuts the thermal photon number at 5 GHz by about eight orders of magnitude, which is why each stage adds attenuation and filtering.

7. Open challenges

- Cryo-CMOS modelling (threshold shift, mismatch, noise) at 4 K
- LO phase noise and DAC resolution limiting gate fidelity
- Thermal anchoring vs. signal integrity of flex cables
- Co-design of qubit chip and control electronics

8. Conclusions

- The bridge, not the qubit alone, limits how far quantum processors scale.
- Each stage balances **heat load**, **noise** and **latency**.
- Near term: multiplexing and better cabling. Long term: cryo-CMOS / SFQ control near the qubits.

References

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